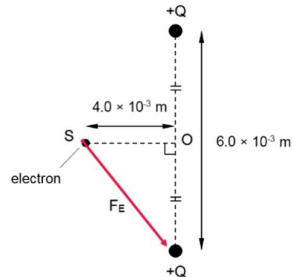


22.

B



$$\theta = \tan^{-1}\left(\frac{4}{3}\right) = 53.13^\circ$$

$$F_E = \frac{Qe}{4\pi\epsilon_0 x^2} = \frac{(1.45 \times 10^{-15})(1.6 \times 10^{-19})}{4\pi\epsilon_0 (5.0 \times 10^{-3})^2} = 8.3444 \times 10^{-20} \text{ N}$$

$$F_{net} = 2 \times F_E \sin \theta = 1.3351 \times 10^{-19} \text{ N}$$

For circular motion,

$$F_{net} = ma_c$$

$$1.3351 \times 10^{-19} = \frac{(9.11 \times 10^{-31})v^2}{4.0 \times 10^{-3}}$$

$$v = 2.4 \times 10^4 \text{ m s}^{-1}$$

22.

A

COF: $E = IR + Ir$